LAB 7: Fuzzing for Input Validation Bugs

Name :Vishwajeet

SRN :PES1UG24AM456

Section: H

Objective

Explain that the experiment uses fuzz testing and the Hypothesis Python library to identify input-validation bugs.

Tools and technologies

Python

Pytest

Hypothesis

4. Application description
Briefly explain the three functions tested:

`average_marks()` – calculates the average of comma-separated marks.

`parse_srn()` – extracts campus, year and department from an SRN.

`short_name()` – converts a full name into an abbreviated name.

5. Initial application and testing

Explain how the original application behaved.

Include a screenshot of the original application running.

Include the screenshot showing the Hypothesis test failure.

Explain the bug discovered: `parse_srn()` accepted an invalid lowercase suffix.

6. Bug fixing
the changes made:

Added input validation for marks.

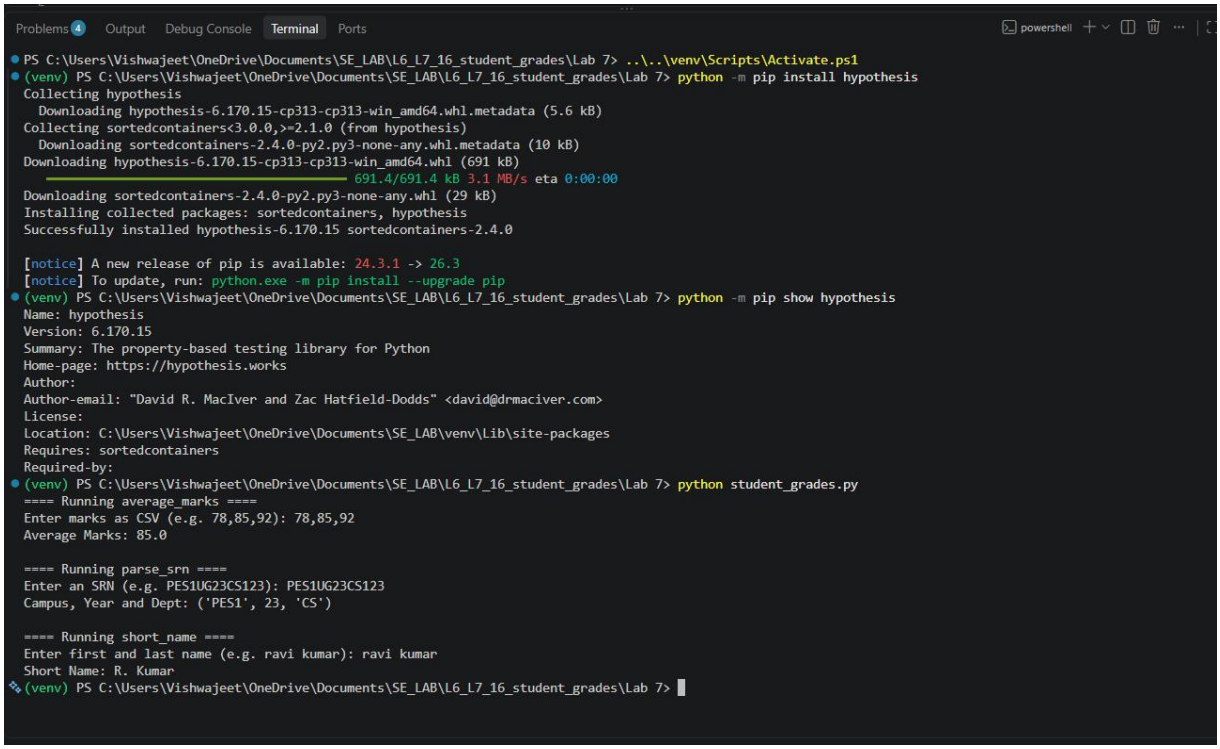
Added strict SRN format validation using regular expressions.

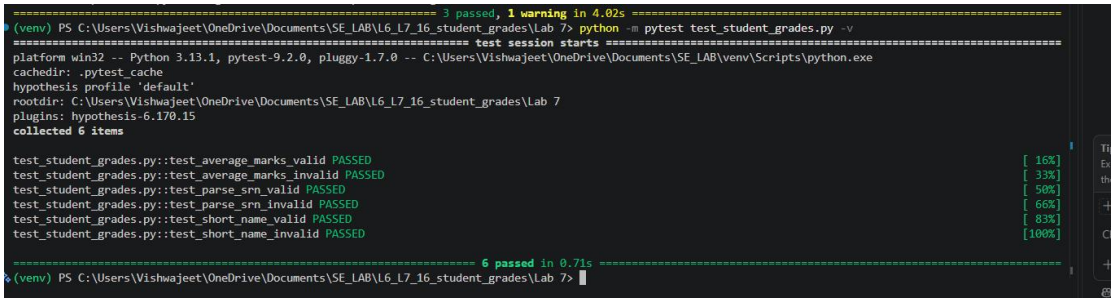
Added validation for names.

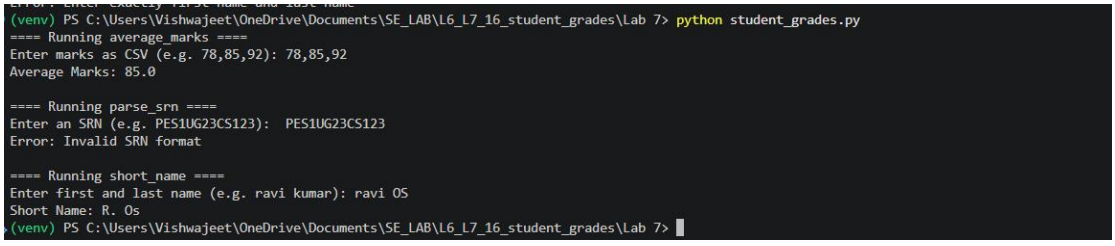
Added appropriate `ValueError` handling.

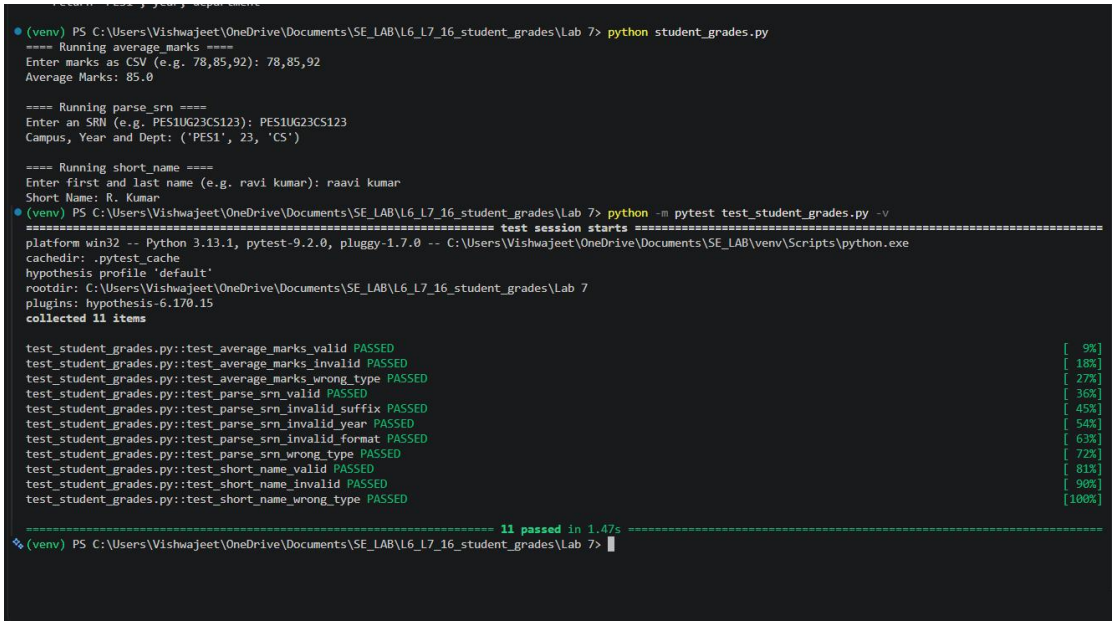
7. Final test results

Include a screenshot showing the final test execution.









8. Conclusion

Explain what you learned about using property-based fuzz testing to discover unexpected inputs and improve input validation.